



Nigeria's sustainable energy future: NECAL 2050 simulations of low carbon pathways

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Received: 19 February 2025 / Accepted: 20 July 2025
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Abstract

Nigeria is dealing with two important issues at the same time: making sure more people have access to energy and lowering greenhouse gas emissions as promised under the Paris Agreement. These promises aim for a 20% reduction in emissions without extra help and a 47% reduction with help by 2030. Finding good ways to cut emissions while still growing the economy is key. This study uses a model called NECAL 2050, which is a version of the UK 2050 Calculator tailored for Nigeria. It looks at seven different future situations, including one that follows the current path and six that focus on cutting emissions with different levels of effort and use of technology. The model takes into account energy sources, how much energy is needed, land use, emissions, and air quality. The results show that without changes, emissions could go up to 604. 11 million tons of CO_{2e} by 2050. However, Scenario 2 and Scenario 4 show big cuts, bringing emissions down to 277. 42 and 274. 60 million tons respectively. These scenarios also cut air pollution by 75% and 76% compared to the current level. These paths focus on using a lot of renewable energy, improving energy efficiency, using bioenergy widely, and using nuclear power in a planned way, while still using fossil fuels in a smaller way. Scenario 2 and 4 are the best options for Nigeria to reach its climate goals and improve energy security. Policymakers should choose these paths by focusing on renewable energy, efficiency, and smart use of nuclear power to make the energy system more sustainable and strong.

Keywords NECAL 2050 · Sustainable energy · Climate change · GHGs emissions and fossil fuels

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1 Introduction

Nigeria is the most populous country in Africa with an estimated population of over 211 million people as of 2021 (Ugwoke et al., 2020). According to the World Bank, approximately 55% of Nigerian population are connected to the grid (World Bank., 2023). Still, out of this percentage only 10% could have regular access of electric supply, making those with irregular access to stand around 45% (IEA, 2023 and SE4ALL., 2023). Worst still, the little energy available in the country depends heavily on fossil fuels as the source, despite the associated environmental, health and socioeconomic impacts (Yusuf et al., 2023). In addition, many people live without even the off-grid electricity to cater for the basic needs (World Bank, 2021 and Edomah et al., 2021). This affects lives as well as the overall country's economic prospects and the standards of living and will make it difficult for Nigeria to achieve the SDGs because all the goals are energy dependent.

The country's economy strongly depends on crude oil and natural gas exports revenue (IEA, 2021). It is also vulnerable to climate change impacts and therefore it has to scale up its climate action to align with the Paris agreement and get out of the challenges posed by the climate change (Climate transparency, 2020). According to a study by the World Health Organization in 2013, the deaths (mostly from women and children) caused by smoke inhalation from firewood usage alone have reached 98,000 per year. Nigeria's total installed electricity generation capacity as at 2023, stands at 13,642 MW. Out of this, natural gas makes up the largest share (74.40%), followed by hydropower (14.7%) and lastly renewable energy- primarily solar PV (11.0%) (IEA, 2023). The country has diverse and abundant energy potentials; including wind energy, Biomass energy, hydroelectric generation, Solar PV and solar thermal as well as nuclear energy (Yusuf et al., 2023; Nyeche & Diemuodeke, 2020; Bamisile et al., 2020a, b; Ewim et al., 2021; Galadima & Aminu, 2020). There is need to harness these resources and chart a new energy future for the country. "Luckily enough", the energy infrastructure of the country is still evolving, therefore the country's energy future depends on the choices it made, and should recognize the need for GHGs mitigation while ensuring that its energy potential is exploited fully (Acikkalp et al., 2014; Akuru et al., 2017).

In the past, the Nigerian government dealt with the problem of not having enough electricity by building more gas-powered plants to generate power. However, the Energy Commission of Nigeria (ECN) wants to use other types of energy as well. To help plan for the future, the ECN did a study to look at how much energy will be needed and how much will be available in different situations. They used a tool called MESSAGE to do this study, and the findings are now part of the current National Energy Master Plan (ECN, 2023). Despite existing studies analyzing Nigeria's energy transition, few have simultaneously assessed the implications of integrated energy pathways on both greenhouse gas emissions and air quality using dynamic systems modeling. This study contributes to filling this gap by using a scenario-based simulation with NECAL 2050 tailored to Nigeria's socio-technical context and energy parameters. It offers practical insights into how renewable energy, energy efficiency, nuclear power, and bioenergy can be combined to meet emission targets without compromising energy access and economic growth. NECAL 2050 is capable of handling the complexity of balancing energy demand, energy supply, and climate mitigation efforts in Nigeria. The tool also has the capacity to integrate country-specific parameters and simulate energy-climate interactions over a multi-decade horizon. The tool was adopted and adapted

by the energy commission of Nigeria, from the UK 2050 Calculator to enable stakeholders to simulate future pathways and explore trade-offs between various energy strategies within a Nigerian context.

Sustainable energy means providing affordable and enough energy for everyone, both now and in the future, in a way that doesn't harm the environment and is acceptable to people and communities. Even though producing energy in developing countries is difficult, it is still a key part of sustainable development because it helps balance economic and social progress with protecting the environment (Khan, 2019). Using energy can cause problems like climate change and environmental damage, such as deforestation, ozone layer loss, air pollution, and the release of harmful substances. These issues need to be solved if Nigeria is to develop in a sustainable way. Nigeria has already agreed to the Paris Agreement, which aims to keep the rise in global temperatures well below 2 °C and to try to limit it to 1.5 °C (Climate Transparency, 2020). The Nigerian government has put in place policies and programs to help develop renewable energy so that it can meet its promises under the Paris Agreement. Some of these policies include the Renewable Energy Master Plan 2015, the National Renewable Energy and Energy Efficiency Policy (2015), Net Metering Regulations (2016), Renewable Energy Feed-in Tariffs (2021), tax breaks and import duty exemptions, the Rural Electrification Agency's Solar Home Systems Program (2022), and the Renewable Energy Transition Assistance Mechanism (2022). However, it is still not clear how Nigeria can clearly and realistically meet its commitments under this agreement.

This study employed the NECAL 2050 model to develop and assess six low-carbon development scenarios in comparison with a reference case. The study evaluates each scenario's effectiveness in reducing greenhouse gas emissions and improving air quality while ensuring continued energy access and economic growth. It provides evidence-based insights to guide policy decisions for Nigeria's sustainable energy transition.

The remainder of this paper is structured as follows. Section 2 provides an overview of the NECAL 2050 model, including its origin, customization for Nigeria, and key assumptions. Section 3 describes the methodology adopted for scenario design and simulation. Section 4 presents and discusses the results of the modeled scenarios, including greenhouse gas emissions, air pollutant levels, and sectoral trends. Section 5 outlines the policy implications derived from the findings. Finally, Sect. 6 concludes with a summary of insights and recommendations for future research and policy direction.

2 Literature review

Many studies have looked at Nigeria's energy system, especially how much energy is needed, how it is provided, or how to reduce greenhouse gas emissions. These studies use different tools and methods. One example is the Energy Commission of Nigeria, which used the MESSAGE model to predict future energy needs and supplies. This information was included in the National Energy Master Plan (ECN, 2023). A study by Michael et al., 2019 simulated low-carbon scenarios using the NECAL 2050 model, while (Kim et al., 2021) employed MESSAGE to evaluate two energy supply pathways for future demand. The SIMPACTS tool was also applied to estimate environmental and damage costs for different electricity generation options (Kim et al., 2021). Similarly, international studies such as those in Sierra Leone (Conteh et al., 2023) and Tamil Nadu, India (Sundararagavan et

al., 2022), have used LEAP and other models to assess renewable energy integration, CO₂ emissions, and supply-demand forecasts.

While these studies have contributed important insights into Nigeria's energy planning, most of them focus on either the supply side or the demand side in isolation, or omit critical co-benefits such as air quality improvements. Moreover, only a few studies have applied the NECAL 2050 model within Nigeria's socio-technical and policy context, and none to our knowledge have used it to evaluate integrated supply-demand scenarios that address both GHG mitigation and air pollution across a multi-decade horizon. There remains a lack of comprehensive, scenario-based studies that explore the synergistic roles of renewables, energy efficiency, bioenergy, and nuclear energy in supporting Nigeria's net-zero ambition.

To assess Nigeria's future energy scenarios, Energy demand was assessed across sectors—transport, industry, buildings, and agriculture—based on population and economic growth projections, while supply calculations are grounded in technology options, fuel availability, and conversion efficiencies.

Despite the growing body of literature on energy transition modeling in Nigeria, most existing studies rely on global or regionally calibrated integrated assessment models that are often opaque, limited in accessibility, or not tailored to Nigeria's specific socio-economic and policy context. Few studies provide a transparent, participatory simulation framework that stakeholders can interact with for long-term energy planning. This study fills that gap by employing the NECAL 2050 model—an open-source, transparent tool adapted specifically for Nigeria—to simulate multiple mitigation scenarios and derive practical policy insights aligned with national climate goals.

2.1 Energy demand and GHGs emissions calculations

To create a good plan for dealing with climate change in the future, we need to figure out how much energy will be needed and how much greenhouse gas emissions will be produced in the future. To do this, it's helpful to build a model of the energy system that can show, at any moment, how much energy is being supplied and how much is being used. According to the ECN report from 2015, this approach is useful. Suppose that:

D_i = Sum of energy demand by a sector in year i .

S_i = Sum of energy supplies by energy form in year i .

L_i = Sum of energy conversion and distribution losses

$$\text{Net supply } (X_i) = S_i - L_i \quad (1)$$

That means, the energy difference (B_i) between the energy demand and supply that will have to be balanced through import, fuels and/or self-generation of electricity is also given by (ECN report, 2015)

$$B_i = D_i - X_i \quad (2)$$

The energy demand was then calculated for different sectors as follows:

2.2 Residential and service sectors

These sectors consist of (i) energy demand and technology pathway (lighting, appliances and cooking energy demand) and (ii) cooling energy demand per house hold and efficiency of cooling systems.

Cooling demand, lighting and appliances demand in these sectors has only electricity as the energy vector while cooking has four vectors namely; electricity, liquid hydrocarbon, gaseous hydrocarbons & traditional fuel (e.g. fire wood, crop residue and charcoal) (ECN report, 2015).

The energy demand for these sectors was calculated by computing the energy demand by vector, the total energy demand by vector, emission by energy vectors and total emissions as per the following equations (ECN report, 2015).

$$LAED = \frac{LAED}{HH} \times HH_n \quad (3)$$

Where; LAED/HH is the energy demand for lightning and appliances per household and HH_n is the number of households.

The cooling energy demand per household is given by (ECN report, 2015):

$$C_kED = \frac{C_kED}{HH} \times HH_n \quad (4)$$

Where; C_kED/HH is the cooking energy demand per household and HH_n is the number of households.

The total energy demand by vector for lighting, appliances and cooking ($TED_{LA\&C}$) is given by (ECN report, 2015):

$$TED_{LA\&C} = LAED + C_kED \quad (5)$$

The GHGs emissions was computed using (ECN report, 2015):

$$X = TED_{LA\&C} \times EF \quad (6)$$

Where: X is the gas emissions (e.g. CO₂, CH₄, N₂O) and EF=emission factor for energy vector.

The cooling energy demand per house hold (CED/HH) was calculated using (ECN report, 2015):

$$CED = \frac{CED}{HH} \times HH_n \quad (7)$$

Finally, the total energy demand by vector in a residential sector was computed using (ECN report, 2015):

$$TED = TED_{LA\&C} + CED \quad (8)$$

2.3 Industrial sector

This sector energy demand was modelled by adding the energy demand in the three industrial subsectors (manufacturing, construction and mining). The Industrial process has two components or trajectories which are growth in Industry with GDP and energy intensity. The energy vectors for this sector Include; Electricity, solid hydrocarbon (e.g. coal), liquid hydrocarbon e.g. fuel oil), gaseous hydro carbon (e.g. natural gas) and traditional fuel (fuel wood) (ECN report, 2015).

The energy demand, process emission and combustion emissions by industry category was computed using the following equation (ECN report, 2015):

$$TED = ED_{man} + ED_c + ED_m \quad (9)$$

Where: ED_{man} , ED_c and ED_m are the energy demand in manufacturing, construction and mining respectively; and TED is the total energy demand.

The energy demand by sub sector (ED) was calculated using (ECN report, 2015):

$$ED = OI \times K \quad (10)$$

Where; OI is the output index, L is the split (share) by vector and K is the energy demand multiplier. The demand by vector (ED_v) was also computed using (ECN report, 2015):

$$ED_v = TED \times L \quad (11)$$

The process emission for a particular greenhouse gas (GHG) was calculated using (ECN report, 2015):

$$GHG = PEG \times N \quad (12)$$

Where PEG is the process emission growth and N is the baseline GHG emission.

Finally, the combustion emission (CE) was calculated using (ECN report, 2015):

$$CE = ED_v \times EF \quad (13)$$

Where EF is the emission factor.

2.4 Transport sector

The mode of transport/technologies considered include car, bus, bike, motorcycle/tricycle, train, air and boat.

The total passenger-km by mode was computed using (ECN report, 2015):

$$Total\ distance\ travel = distance\ travel\ per\ person \times population \quad (14)$$

$$Mode\ share = \frac{distance\ traveled\ by\ mode}{total\ distance\ travelled} \times 100 \quad (15)$$

The passenger-km by mode/technology was computed using (ECN report, 2015):

$$Ownership = \frac{EquationNumber \text{ of vehicles}}{population} \quad (16)$$

$$Occupancy = \frac{EquationNumber \text{ of persons}}{population} \quad (17)$$

$$Technology \text{ penetration} = \frac{technology}{\sum technology \text{ in mode}} \times 100 \quad (18)$$

The energy used by mode and vector for all the transport sectors was computed using (ECN report, 2015):

$$Energy \text{ demand} = \sum distance \text{ travelled by mode \& technology} \times efficiency \text{ of technology} \quad (19)$$

$$Efficiency \text{ of technology} = \frac{petrol \text{ (litre)}}{distnace \text{ travelled (km)}} \quad (20)$$

The emission by mode was computed using (ECN report, 2015):

$$Emission = \sum distance \text{ travelled by technology} \times emission \text{ factor}$$

$$Emission \text{ factor} = \frac{CO_2 \text{ (kg)}}{distance \text{ traavelled (km)}} \quad (21)$$

2.5 Energy supply calculations

The procedures applied for the calculation of the electricity supply via gas power plant, self-generation (oil), coal power plant, biomass, renewables and nuclear power plants are as follows (ECN report, 2015):

$$Available \text{ generation} = Available \text{ capacity} \times load \text{ factor} \quad (22)$$

$$Total \text{ generation} = Available \text{ generation} + plant' \text{ s own requirement} \quad (23)$$

$$Resource \text{ requirement} = \frac{Total \text{ generation}}{Thermal \text{ efficiency of plant}} \quad (24)$$

$$GHG \text{ emission} = resource \text{ requirement} \times emission \text{ factor} \quad (25)$$

3 Methodology

This section describes the methodological approach used to simulate Nigeria's low-carbon energy transition scenarios. It outlines the design of seven scenarios, including a reference case and six mitigation pathways, as well as the assumptions and parameterization processes employed within the NECAL 2050 framework. The section also explains how various energy, emissions, and land-use levers were adjusted to evaluate potential outcomes across key sectors.

3.1 About the Nigeria energy calculator (NECAL 2050)

NECAL 2050 is an energy and emissions model in form of a software developed by the Energy Commission of Nigeria based on the methodology used in the development of the UK 2050 pathways calculator. It was developed with the support of department of energy, UK and the British High Commission in Nigeria in order to assist Nigeria to achieve its climate goals. The software is an energy planning tool that gives the energy demands and supply pathways for different periods and their consequent air quality and greenhouse gas emissions. It is a practical tool for exploring the options by which Nigeria can sustainably develop its energy sector. NECAL 2050 is an open source integrated model of energy, emissions and land use in Nigeria that is aimed at identifying energy secure pathways for supply and demand of energy between now and 2050 (INDC 2015). The software allows users to develop their peculiar pathway combinations in an attempt to achieve emissions reduction and ensure energy security based on available technologies, resources and behavioral changes. However, due to the complex interaction between energy modelling and other domains (e.g. socioeconomic, environmental and technical aspects), the software was developed using scenario based models involving many assumptions. That is why the software outputs are used as a demonstration of the scale of changes required and the choices which are likely available. NECAL 2050 has a range of four trajectories as depicted in Fig. 1 for the scale of changes that might occur (ECN report, 2015). The trajectories reflect the range of potential future scenarios likely to be seen in a particular sector. The software is not a cost optimization tool, but considers only what is deliverable and the least cost pathway in each sector within the period (2010–2050) under different considerations and assumptions.

NECAL 2050 enables users to understand the possible scenarios and their implications in order to contribute to the sustainable energy development in Nigeria. The structure of the software is presented in Fig. 2, moreover detailed methodology including the models and assumptions (technical, economic, environmental and societal) used by the software in the calculation of; energy demand, energy supply, GHGs emissions and air quality is available in (ECN report, 2015 and INDC, 2015). Furthermore, the different trajectory choices and their interpretations, the energy supply and demand vectors as well as the detailed global assumptions are also in (ECN report, 2015 and INDC, 2015).

While the NECAL 2050 model provides a robust framework for scenario-based energy planning, it has limitations. First, it is a deterministic calculator that does not simulate uncertainty or stochastic variability. Second, it does not model price dynamics, international trade, or geopolitical influences. Third, technology adoption is modeled through fixed pathways without accounting for behavioral inertia or institutional resistance. Additionally, the model simplifies sectoral interactions, which may underestimate system-wide feedback

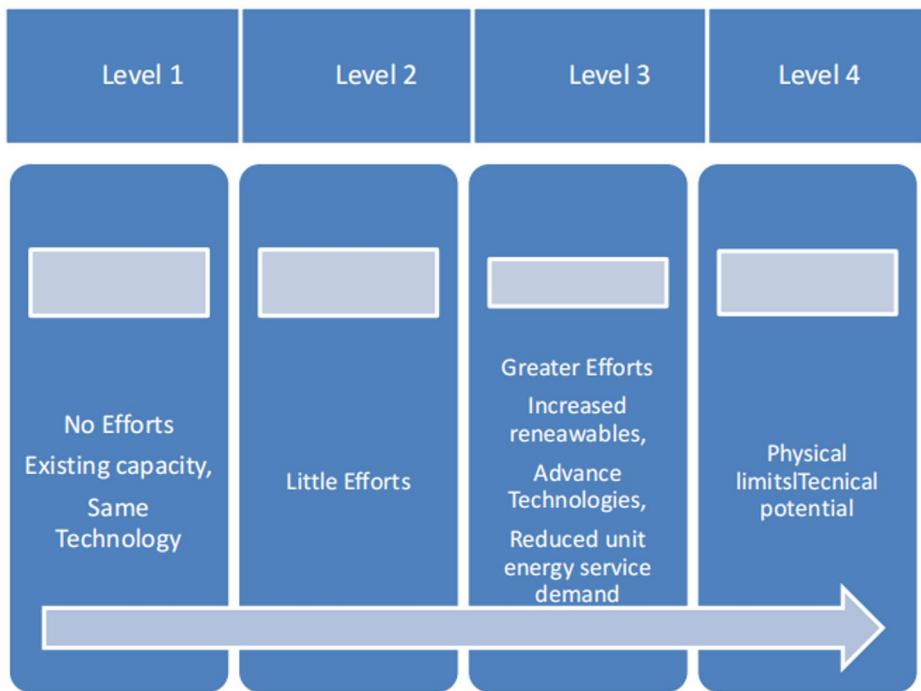


Fig. 1 Levels/Trajectories Settings (ECN report, 2015)

effects. Despite these constraints, the model remains valuable for high-level policy analysis and comparative scenario evaluation.

3.2 Scenarios development

3.2.1 Considerations for the scenarios development

The choice of penetration levels of nuclear energy, renewables, energy efficiency, and bio-energy in each scenario were informed by existing national policy documents, including the “Nigeria Renewable Energy Master Plan”, “the Energy Transition Plan”, and the “Long-Term Low Emission Development Strategy”. Scenario levels were also aligned with infrastructure availability, investment feasibility, and political acceptability. Furthermore, the following documents were also considered in the development of the scenarios:

- I. The Energy profile of Nigeria’s indicated country’s key mitigation measures to meet its energy intended nationally determined contributions (ECN, 2023).
- II. The Nigeria energy compact template which contains the following 2030 targets on how Nigeria can achieve SDG 6 targets.
- III. The identified actions that Nigeria can do for sustainable energy development as documented by the National renewable and energy efficiency policy and the Nigeria nationally determined contributions (NREEEP, 2014).

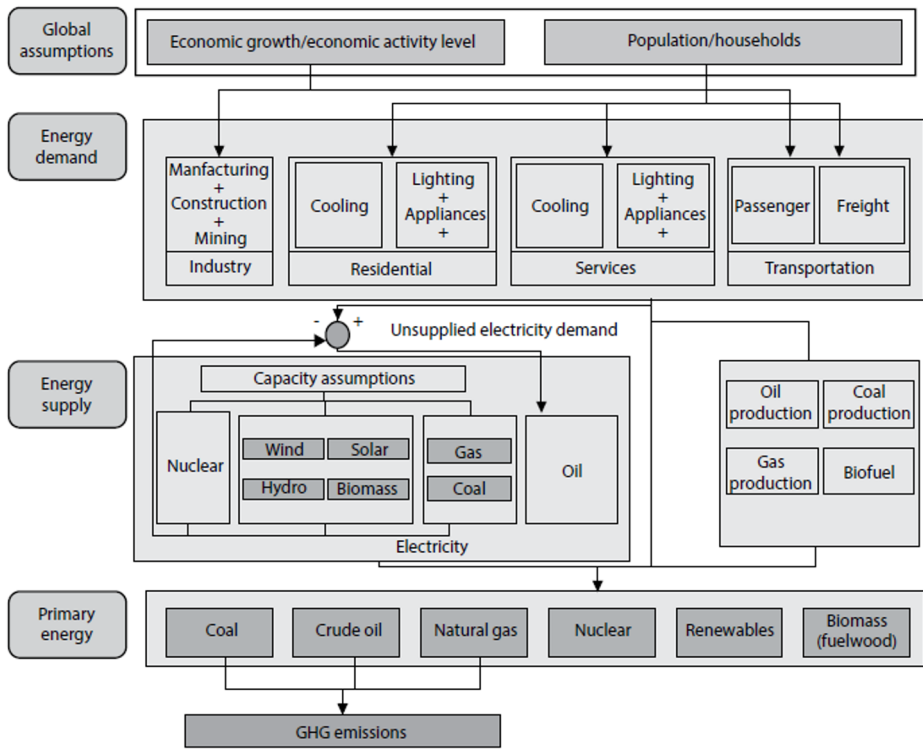


Fig. 2 Structure of NECAL 2050 (ECN report, 2015)

The NECAL 2050 model was built using data specific to Nigeria, including information from national statistics, energy balance sheets, and environmental reports. Important inputs came from the National Population Commission for population predictions, trends in energy use across different sectors, local factors that determine emissions from fossil fuels, and how much of the population has access to electricity. To make the model fit Nigeria, it used data on population growth, how quickly cities are expanding, future energy needs, and emissions factors for different sectors. The model made some key assumptions, like how quickly new technologies will be adopted, how much land will be available for use, and how much fuel will need to be imported. These assumptions about future technology use, energy efficiency, and industrial production were based on the National Development Plan and discussions with experts during workshops held by the Energy Commission of Nigeria and GIZ.

3.2.2 The developed scenarios

Figure 3 illustrates the simulation workflow, showing the transformation of input levers into quantitative outputs such as emissions trajectories, energy supply-demand balances, and sectoral contributions. Six scenarios were developed (listed as SN1, SN2, SN3...SN6 below) as reasonable and practicable as possible based on the foregoing considerations in addition to the business as usual scenario. Renewables, nuclear, energy efficiency and bio-energy were set at different levels. Level 1 is the low level (referred to as the business

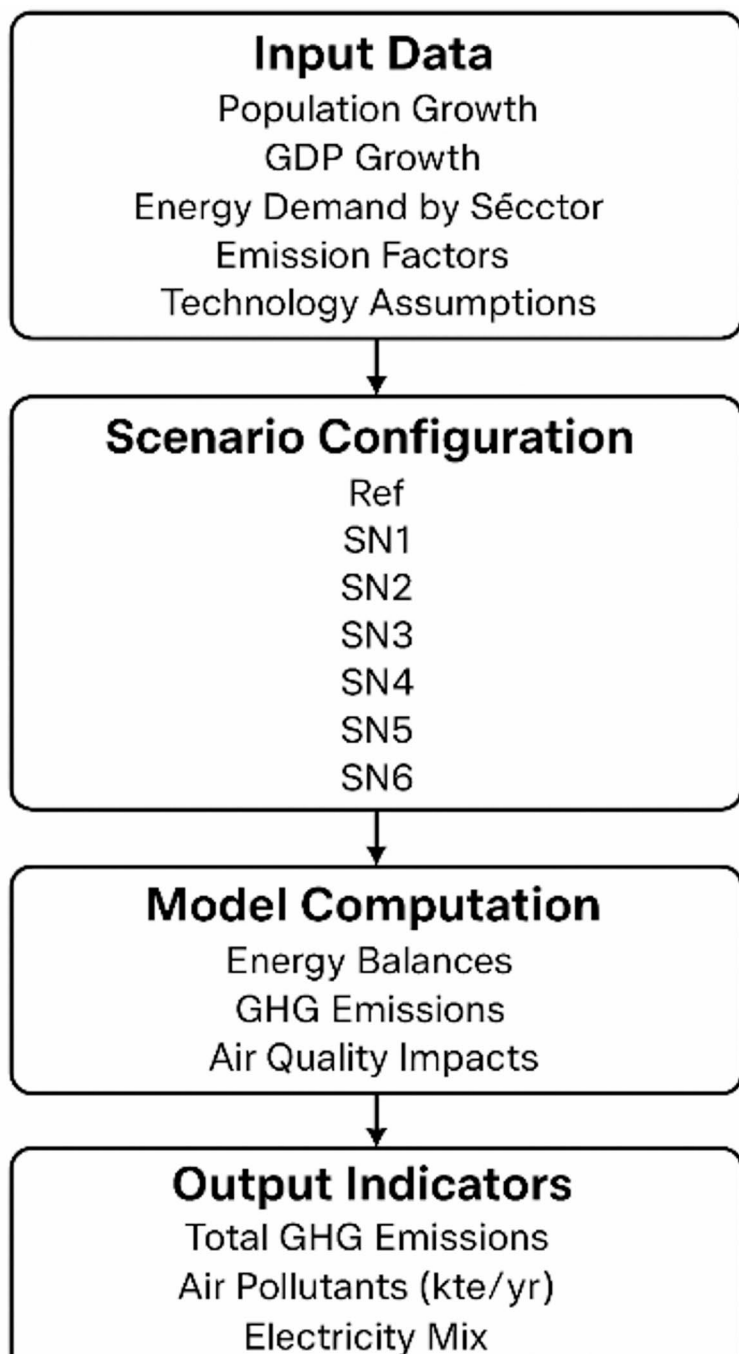


Fig. 3 Simulation workflow showing the transformation of input levers into quantitative outputs

as usual), moderate level, high level and very high level. The scenarios developed were (Table 1):

- I. Reference scenario (Ref SN): This refers to a situation where by Nigerian government does no action to mitigate GHGs emission, improve energy supply and reduce energy

Table 1 Levels attached for each of the developed scenarios in this work

		Final energy demand (PJ/year)						
		Ref SN	SN 1	SN 2	SN 3	SN 4	SN 5	SN 6
Transport	1. Domestic transport behaviour	1	2	4	3	4	2	1
	2. shift to low carbon emission transport technology	1	2	4	3	4	2	1
	3. Fuel switch for internal combustion engine	1	2	4	3	4	2	1
	4. Domestic freight	1	2	4	3	4	2	1
	5. Cooling demand	1	2	4	3	4	2	1
	6. Efficiency of cooling system	1	2	4	3	4	2	1
Households	7. Residential lighting, appliances and cooking	1	2	4	3	4	2	1
Industry	8. Technology pathway (cooking)	1	2	4	3	4	2	1
	9. Growth in industry with GDP	1	2	4	3	4	2	1
	10. Energy intensity of industry	1	2	4	3	4	2	1
Services/commercial	11. Service sector demand for cooking	1	2	4	3	4	2	1
	12. Efficiency of cooling system	1	2	4	3	4	2	1
	13. Service sector lighting, appliances and cooking	1	2	4	3	4	2	1
	14. Technology pathway (cooking)	1	2	4	3	4	2	1
Nigeria electricity generation	Primary energy supply (Pj/year)							
	1. Natural gas power station	1	3	3	3	3	3	3
	2. Biomass power	1	2	4	3	4	2	1
	3. Coal power stations	1	2	2	2	2	2	2
	4. Nuclear power stations	1	4	1	3	2	1	4
	5. Wind	1	4	4	3	4	2	1
	6. Hydroelectric power stations	1	4	4	3	4	2	1
	7. Small hydroelectric power stations	1	4	4	3	4	2	1
	8. Grid connected solar PV	1	4	4	3	4	2	1
	9. Concentrated Solar power	1	4	4	3	4	2	1
	10. Standalone solar PV	1	4	4	3	4	2	1
	11. Land dedicated to bioenergy	1	4	4	3	4	2	1
	12. Livestock and their management	1	4	4	3	4	2	1
	13. Volume of waste and recycling	1	4	4	3	4	2	1
	14. Type of fuels from biomass	1	4	4	3	4	2	1
	15. Bioenergy imports	1	1	1	1	1	1	1
	16. Electricity imports	1	1	1	1	1	1	1
Nigeria fossil fuel production	Greenhouse gas emissions (Mt CO ₂ e/year)							
	1. Indigenous fossil fuel: coal	1	2	2	3	3	2	1
	2. Indigenous fossil fuel: Oil	1	3	2	2	3	4	4
	3. Indigenous fossil fuel: Gas	1	4	4	4	3	4	4

demand up to 2050. This is to enable comparison with the other developed scenarios in order to study the implication of the proposed pathway on the energy demand, energy supply, GHGs emission and air quality.

- II. Scenario 1 (SN1): Very high penetration of renewables and nuclear energy with moderate energy efficiency and bioenergy.
- III. Scenario 2 (SN2): Very high renewables, energy efficiency and Bioenergy with low nuclear penetration.
- IV. Scenario 3 (SN3): High renewable, nuclear, Bioenergy and energy efficiency.
- V. Scenario 4 (SN4): Moderate nuclear; with very high renewables, bioenergy and energy efficiency.
- VI. Scenario 5 (SN5): Low nuclear; with moderate renewables, energy efficiency and bioenergy.
- VII. Scenario 6 (SN6): Highest nuclear; Low renewables, bioenergy and energy efficiency.

4 Results and discussion

This section presents and analyzes the outcomes of the scenario simulations. It discusses the projected trends in GHG emissions, air pollutant levels, and sectoral transformations across different pathways.

4.1 Total final energy demand and supply

The total final energy demand by sectors in the year 2050 for the considered scenarios were presented in Table 2. The total final energy demand (Twh) in the year 2050 for the considered energy vectors (transport, industry, cooling, lighting & appliances and cooking) was estimated at 4352.67 Twh under the reference scenario (business as usual case) and, 2776.30 Twh, 1454.45 Twh, 2031.19 Twh, 1448.06 Twh, 2773.12 Twh and 4341.60 Twh under scenario 1, 2, 3 4, 5 and 6 respectively. It was found that the energy demand decreased from the business as usual case by; 36% under SN1, 67% under SN2, 53% under SN3, 67% under SN4, 36% under SN5 and 0.3% under SN6. The least decrease in the final energy demand and supply was found to be associated with scenario 2 and 4 and they were depicted in Figs. 1 and 2 respectively.

The total final energy demand for the reference scenario was found to be 2–3 times that of the other simulated scenarios (SN1 –SN5) because in the other simulated scenarios some of efforts related to energy efficiency improvement, renewable and bioenergy were introduced unlike the business as usual case. Specifically, there has been an assumption that moderate to high energy efficiency levels will be achieved by 2050. However, the total final

Table 2 Final energy demand (Twh) by sectors in the year 2050 for the scenarios considered

Vector	Ref SN	SN 1	SN 2	SN 3	SN 4	SN 5	SN 6
Transport	686.85	365.15	189.32	242.12	189.32	365.15	686.85
Industry	1389.83	712.02	282.58	430.94	276.19	708.84	1378.76
Cooling	487.75	351.18	136.57	234.12	136.57	351.18	487.76
Lighting & appliances	1200.48	900.36	678.72	816.58	678.72	900.36	1200.48
Cooking	587.75	447.59	167.27	307.43	167.27	447.59	587.75
Total	4352.67	2776.30	1454.44	2031.19	1448.06	2773.12	4341.60

energy demand for scenario 6 is very close to that of the reference scenario which indicates that it is not significantly different from the business as usual case ($P < 0.05$). This is due to the assumption that very high nuclear energy will be introduced into the country without any change in the levels of renewable energy generation, bioenergy production and energy efficiency. This unequivocally confirmed that deployment of nuclear energy alone cannot cater for the energy demand of Nigeria but it should be done together with other renewables.

These results demonstrate the central theoretical principle of integrated energy system transitions: improvements in energy efficiency—when combined with renewables and bioenergy—can drastically reduce energy demand, even under growing population and economic development pressures. Scenarios 2 and 4, which incorporate very high levels of energy efficiency, achieved the most significant reductions, consistent with global energy transition theory that links end-use efficiency improvements to system-wide demand reductions.

In contrast, Scenario 6, which introduces high nuclear energy without improvements in demand-side efficiency or diversification, shows little deviation from the baseline. This outcome underscores a key insight in energy transition literature: supply-side interventions alone, particularly those that do not alter consumption behaviour or system efficiency, often yield limited impacts on overall demand.

Furthermore, the findings reaffirm the low-carbon development theory, which suggests that cost-effective emissions reductions in developing countries require simultaneous changes across the energy supply and demand spectrum. This has practical implications not just for Nigeria, but also for other developing economies with evolving infrastructures—where early investment in energy-efficient systems can lead to long-term benefits in sustainability, affordability, and resilience.

Meanwhile, the primary energy supply in the year 2050 based on the considered energy supply vectors (Nuclear, solar, wind, hydro, coal, oil and gas) and scenarios (SN1–SN6) was presented in Table 3. Coal, oil and gas are not clean energy sources but were included in the simulation because they are highly abundant in Nigeria and cannot be left off during the energy transition planning as they are a major source of revenue to the country without which there may be no resources for the transitioning. The total primary energy supply for the reference scenario was 5901.92 Twh which is greater than the total primary energy supply for scenarios 1 to 5 (4405.94 Twh for scenario 1; 2664.34 Twh for scenario 2; 4050.94 Twh for scenario 3; 2720.93 Twh for scenario 4 and 4426.4 Twh for scenario 5). However, scenario 6 had a total primary energy supply of 6548.45 Twh which is greater than that of

Table 3 2050 primary energy supply (Twh) in the year 2050 based on the scenarios

Vector	Ref SN	SN 1	SN 2	SN 3	SN 4	SN 5	SN 6
Nuclear	0.00	567.84	0.00	154.28	107.14	0.00	567.84
Solar	8.65	435.67	435.67	131.93	435.67	49.09	8.65
Wind	0.018	22.09	22.09	5.70	22.09	1.75	0.02
Hydro	10.27	74.51	74.51	51.01	74.51	29.88	10.27
Electricity oversupply (imports)	−0.53	−0.53	−0.53	−0.53	−5.75	−0.53	−0.53
Bioenergy	1346.47	1684.24	1440.87	1321.18	1440.87	1100.46	1352.32
Coal	−13.46	120.01	320.03	177.42	319.93	49.34	37.00
Oil	3741.27	1145.91	233.04	1058.48	191.76	1952.02	3048.78
Gas	809.26	356.19	138.65	1151.47	134.72	1244.38	1524.08
Total	5901.92	4405.94	2664.34	4050.94	2720.93	4426.4	6548.45

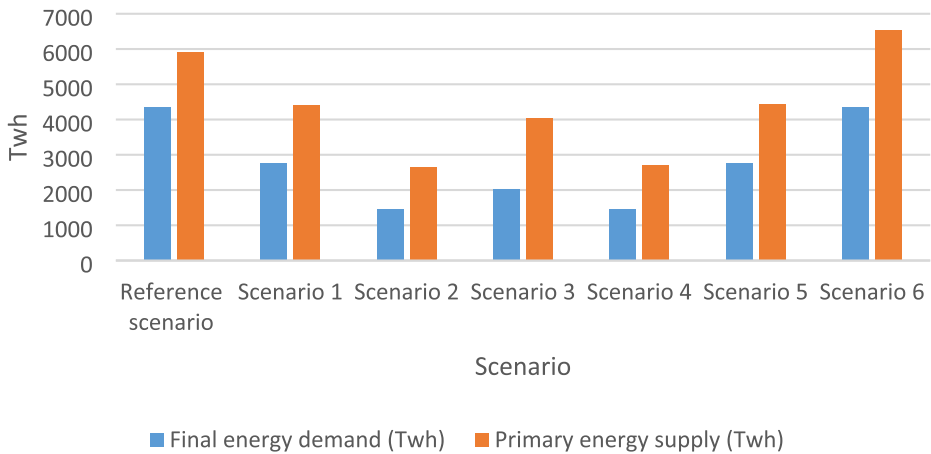


Fig. 4 Total final energy demand and primary energy supply for the year 2050

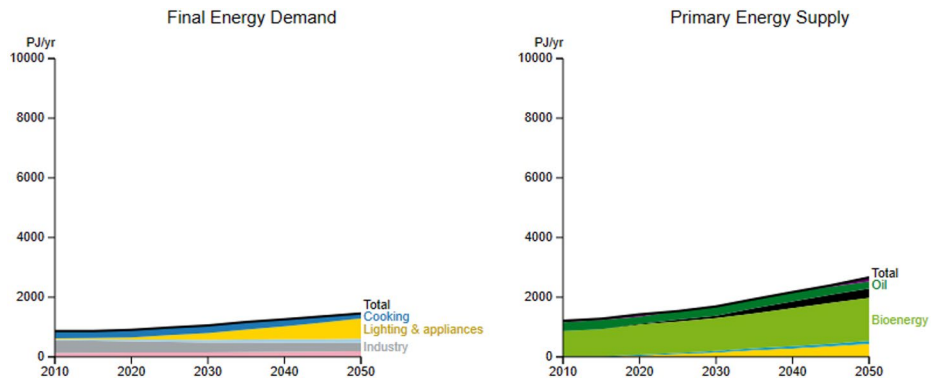


Fig. 5 Scenario 2 final energy demand and supply

the reference scenario by 646.53 Twh. NECAL 2050 always balances energy demand and energy supply as can be seen from Figs. 4, 5 and 6 by ensuring that the final energy supply is equal or greater than the energy demand to ensure energy security. The total primary energy supply was greater than the total energy demand by 26% under the reference scenario and by; 37%, 45%, 50%, 47%, 37% and 34% for scenario 1, 2, 3, 4, 5 and 6 respectively. The total energy supply for the reference scenario was 5901.918 Twh, and decreased by 25% for SN1, 55% for SN2, 31% for SN3, 54% for SN4, 25% for SN5, meanwhile it increased by 11% for SN6. All the hypothetical scenarios except SN6, require a significant change in energy efficiency which depends on technological and behavioural changes. However, scenario 6 provides even greater final energy supply than the reference scenario even though the energy efficiency is maintained at status quo which confirms the significant role that nuclear energy will play in the energy mix.

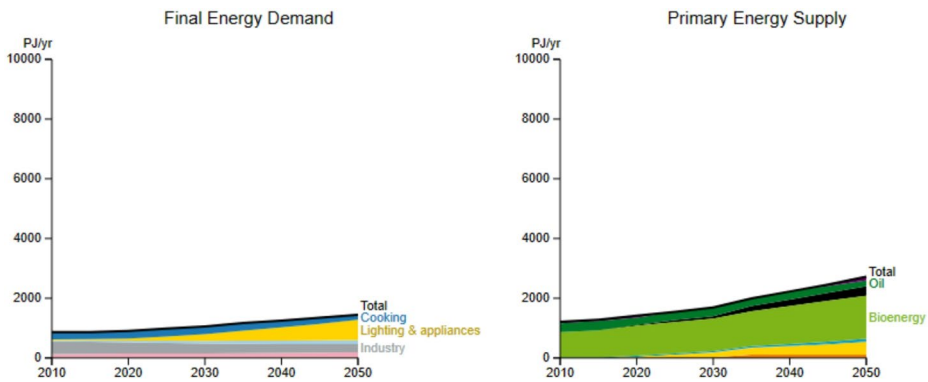


Fig. 6 Scenario 4 final energy demand and supply

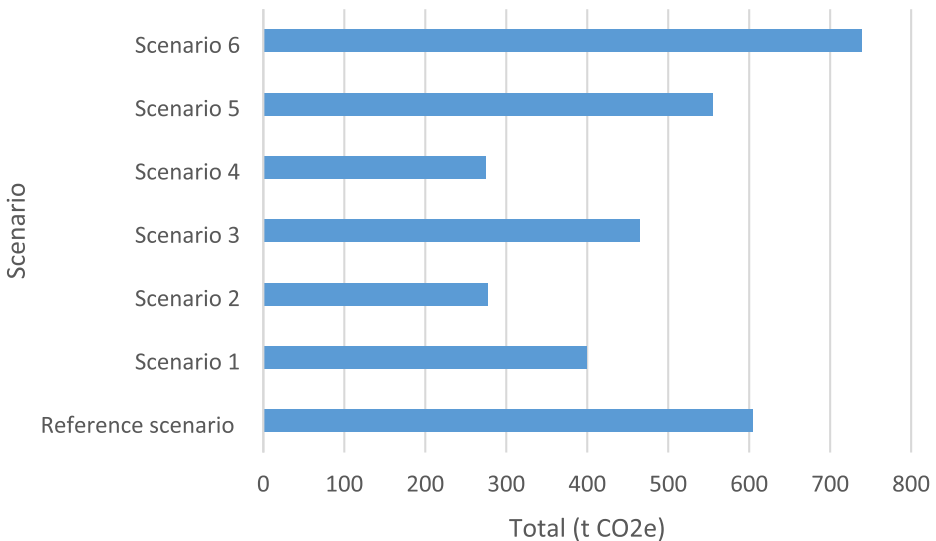


Fig. 7 Totals GHGs emitted under various scenarios

4.2 Greenhouse gas emissions (GHGs)

Nigeria commits to unconditionally reduce emissions by 20% below business as usual by 2030 with conditional increase to 47% provided that sufficient international support is forthcoming. In the light of this commitment, greenhouse gas emissions from the fuel combustion, industrial processes, agriculture, Land Use, Land-Use Change and Forestry (LULUCF), waste and bioenergy credit sectors were added for the considered scenarios. The total GHGs emissions for the reference scenario was estimated at 604.11 t CO₂e and; 399.23 for SN1, 277.42 for SN2, 465.22 for SN3, 274.60 for SN4, 554.87 for SN5 and 739.29 for SN6. Figure 7 shows the total estimated greenhouse emissions in 2050 for the six considered reference scenarios (SN1 to SN6). The total estimated emissions for scenario 1

to scenario 5 is less than that emissions by the reference scenario, while that emitted by the scenario 6 was even greater by 135.18 t CO₂e.

The estimated GHGs emission is in the order; scenario 4 < scenario 2 < scenario 1 < scenario 3 < scenario 5 < reference scenario < scenario 6. The GHGs emission decreased by 34% for SN1, 54% for SN2, 23% for SN3, 55% for SN4, 8% for SN5. However, the GHGs emission even increased by 22% for SN6 which disqualifies this scenario in the light of sustainable development. Scenario 2 and 4 (depicted in Figs. 8 and 9) are the best scenarios for Nigeria to fulfil its Paris agreement. The agreement considered unconditionally reducing emissions by 20% below business as usual by 2030 with conditional increase to 47%. These two scenarios envisage a future with very high penetration of renewables, energy efficiency and Bioenergy with least nuclear to moderate nuclear intervention while still utilizing our fossil fuels (coal, oil and gas) adequately. Scenarios 2 and 4 outperform others due to their balanced approach: they combine aggressive deployment of mature and cost-declining technologies like solar PV, wind, and bioenergy with feasible levels of energy efficiency measures. These options are already gaining social acceptance in Nigeria due to off-grid solar initiatives and rising diesel prices. luckily, both scenarios align with ongoing government initiatives such as the Nigeria Energy Transition Plan and the National Renewable Energy and Energy Efficiency Policy.

Furthermore, the simulation results indicated that taking an uncertainty of 5%, scenario 1 is also a candidate for reducing emissions by 20% even if the international support is not forthcoming. This scenario considers introducing highest renewables and nuclear with moderate energy efficiency and bioenergy. The results obtained could not be compared with that of other similar researches on energy policy modelling such as Michael et al., 2019 and Adeyemi-Kayode et al., 2022, this is because the scenarios are different.

Though the results from Scenario 2 and Scenario 4 are promising, it is worth noting that high renewable and efficiency penetration require substantial grid upgrades, storage

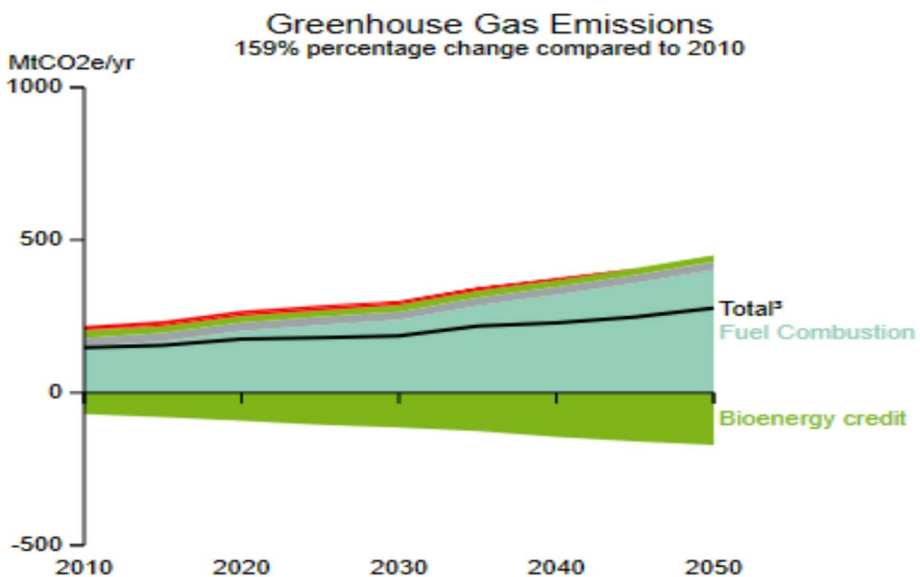


Fig. 8 Scenario 2 based GHGs emissions

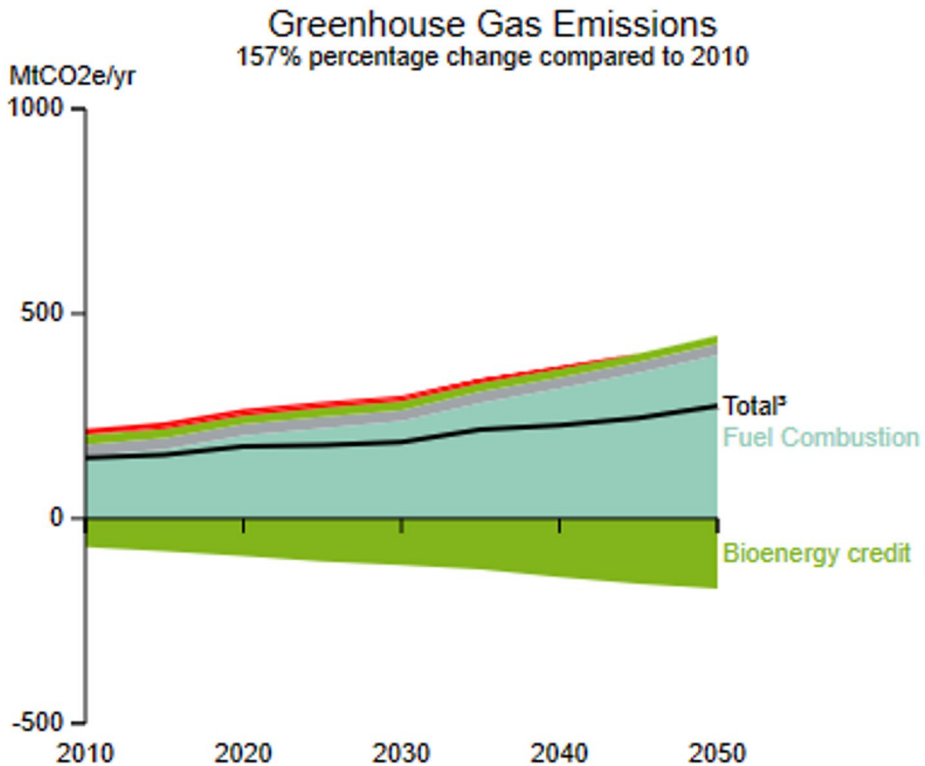


Fig. 9 Scenario 4 based GHGs emissions

systems, and capacity-building. Moreover, investment in these technologies remains constrained by fiscal limitations and global capital access. However, recent government efforts such as; Climate Change Act and the National Energy Transition Plan, the ETP 2060 and the Presidential Power Initiative signal increasing commitment to a sustainable energy transition, suggesting that with international support and domestic policy alignment, these scenarios are attainable.

These results align with the theoretical frameworks of low-carbon transitions in the Global South, which emphasize the importance of technology mix, institutional context, and policy alignment. Scenario 2 and Scenario 4 illustrate how multi-vector strategies—high renewables, strong energy efficiency, moderate nuclear reliance—can outperform narrow, supply-side pathways. This supports the multi-level perspective (MLP) in transition theory, where change arises not only from technological shifts but from system-level co-evolution of institutions, infrastructure, and social acceptance.

Moreover, the performance of Scenario 6, despite its high nuclear content, reinforces concerns in energy transition literature about technological lock-in and high capital intensity of centralized systems. Without corresponding improvements in demand-side efficiency or distributed renewables, centralized nuclear pathways may be insufficient for climate targets—especially in contexts with weak grid infrastructure and financing constraints.

The observed synergy between renewables, efficiency, and bioenergy also reflects the “polycentric approach” to energy governance, where decentralized interventions can be scaled effectively with community support and policy incentives. Indeed, Nigeria’s growing off-grid solar market and bioenergy adoption demonstrate real-world resonance with these pathways.

4.3 Air quality

The estimated kilo tonnes equivalent of air pollutant released per year (kte/yr) for the reference scenario was 4395 kte/yr. The kte/yr decreased from the reference scenario by 56%, 75%, 54%, 76%, 36% and 16% under scenario 1, 2, 3, 4, 5 and 6 respectively. The air quality in 2010 and in the 2050 based on the simulated scenarios were presented in Fig. 10a and b. The air quality under these scenarios as measured by kte/yr was such that in the order of decreasing air quality: Reference scenario < Scenario 6 < scenario 5 < scenario 3 < scenario 1 < scenario 2 < scenario 4. This further confirmed scenario 2 and 4 to be the best scenarios in terms of decreasing health burden associated with pollution from energy generation and utilization. The air pollutants considered in the simulation that contributed were NO_x, SO₂, PM₁₀ and NMVOC and their different level of contributions to be overall pollution is indicated in Fig. 11. Total air pollutants (kte/year) due to PM₁₀, NO_x, SO₂ and NMVOC were also calculated for 2010 and 2050. It was (12442.34 to 4395.44 kte/yr) reference scenario, (12439.12 to 1927.76) SN1, (12417.49 to 1097.30) SN2, (12426.36 to 2035.17) SN3, (12417.49 to 1044.84) SN4, (12439.86 to 2823.61) SN5 and (12442.34 to 3712.90 kte/yr) SN6.

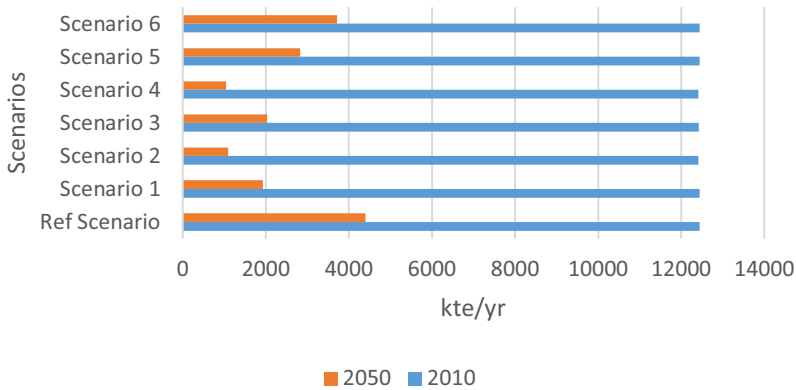
Improved air quality under Scenarios 2 and 4 has significant public health and economic implications. Reduced exposure to PM₁₀ and NO_x emissions could lead to lower rates of respiratory diseases, cardiovascular illness, and premature deaths. According to WHO data, air pollution costs Nigeria over \$2.1 billion annually in healthcare and productivity losses. The estimated 75–76% improvement in air quality could result in substantial national savings and improved quality of life, especially in urban areas.

These results align with the co-benefits framework in climate and public health literature, which asserts that emissions mitigation policies can yield substantial gains in local air quality and public health, especially in low- and middle-income countries. This study adds theoretical value by quantifying these air quality benefits in a West African context using an integrated scenario-based approach, thus contributing to the sparse literature on air-pollution-related health externalities in African energy transitions.

From an environmental justice perspective, the benefits of reduced air pollution are especially important in Nigeria, where vulnerable populations—particularly in dense urban areas—bear the brunt of poor air quality. The findings from Scenarios 2 and 4 suggest that a well-designed low-carbon transition can also serve as a health equity strategy, reducing respiratory illness and lowering economic burdens from pollution-linked diseases. According to the World Health Organization, air pollution costs Nigeria an estimated \$2.1 billion annually in health and productivity losses. Therefore, achieving a 75–76% reduction could result in significant national savings and life expectancy gains.

Conceptually, this reinforces the energy-health-development nexus: energy system decisions are not solely technical or economic, but have profound implications for societal well-being and sustainable development. Integrating air quality as a planning metric alongside

(a)



(b)

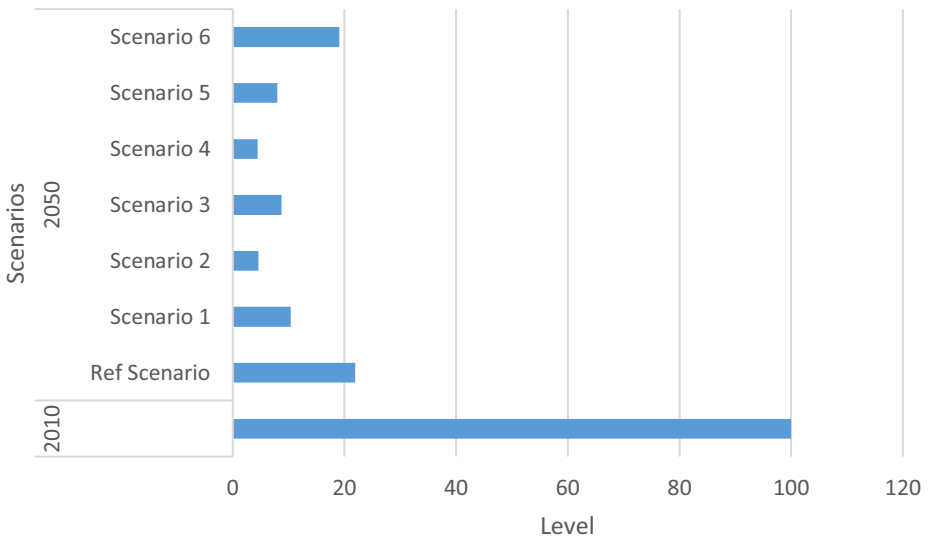


Fig. 10 a: Total air pollutants in the 2010 and 2050. b: Standardized pollutants index in the 2010 and 2050

GHGs aligns this study with the sustainable development co-benefits discourse and supports a holistic planning approach in developing-country energy policy.

5 Policy implications

The simulation results from NECAL 2050 highlight specific pathways and levers that have the most significant impact on achieving Nigeria's emission reduction targets. The following policy recommendations are evidence-based and aligned with the study's scenario outcomes, particularly Scenarios 2 and 4:

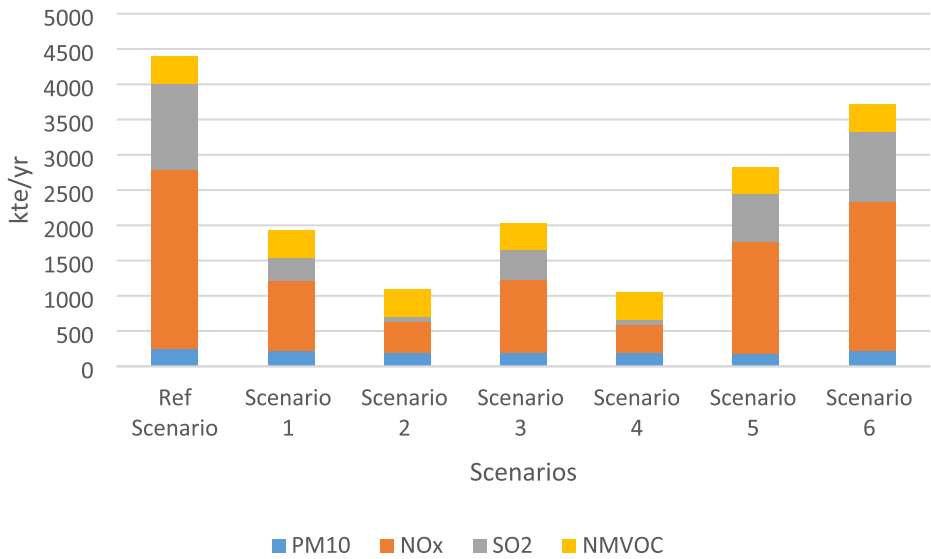


Fig. 11 Total air pollutants released in the year 2050

5.1 Prioritize High-Penetration renewable energy deployment

Given the substantial emission reductions achieved in scenarios with strong solar and wind integration, policymakers should expand renewable energy targets beyond current levels. This includes removing regulatory and financial barriers to utility-scale solar, incentivizing off-grid and mini-grid systems, and integrating renewables into the national grid through investments in storage and smart transmission infrastructure.

5.2 Scale up energy efficiency across End-Use sectors

Both Scenario 2 and 4 show that aggressive energy efficiency measures significantly reduce GHG and air pollutant emissions. Policies should enforce strict efficiency standards in the building, industrial, and transport sectors, promote energy audits, and provide incentives for the adoption of efficient technologies such as LED lighting, improved cook stoves, and electric vehicles.

5.3 Enable strategic and safe nuclear energy integration

A moderate deployment of nuclear energy in Scenarios 2 and 4 contributed to deep decarbonization. Nigeria should finalize its civil nuclear framework, invest in capacity building, and explore international partnerships to responsibly integrate small modular reactors into its energy mix as a stable baseload option to complement renewables.

5.4 Strengthen land use and bioenergy governance

The modeled scenarios with high bioenergy use (especially for cooking and transport) show co-benefits in emissions reduction and air quality. Policymakers should promote sustainable biomass value chains, incentivize second-generation biofuels, and enhance forest management to avoid land-use conflicts and emissions leakage.

6 Conclusion

This study used the NECAL 2050 simulation model to explore Nigeria's potential pathways toward meeting its climate commitments under the Paris Agreement while addressing the dual challenges of energy access and environmental degradation. Among the six scenarios evaluated, Scenarios 2 and 4 emerged as the most effective in reducing greenhouse gas (GHG) emissions—by 54% and 55%, respectively—and in improving air quality—by 75% and 76%, respectively—relative to the reference scenario. Scenario 1, under a 5% uncertainty margin, also qualifies for meeting the unconditional 20% emission reduction target. These results provide clear evidence that a balanced energy mix—featuring high deployment of renewables, energy efficiency measures, and bioenergy, alongside a modest integration of nuclear power—can significantly lower emissions and air pollutants while sustaining energy supply.

The study's contribution to theory lies in its application of an integrated modeling framework to a developing country context, demonstrating the value of systems-based scenario analysis for low-carbon transitions. It also advances the theoretical discourse on co-benefits, showing how climate mitigation strategies can simultaneously deliver substantial public health and economic gains through improved air quality. From a practical perspective, the results offer clear policy guidance: Nigeria's energy transition strategy should prioritize distributed renewables and efficiency improvements, supported by enabling policies and investments in grid infrastructure and clean technology capacity. The findings are particularly relevant to ongoing initiatives such as the Nigeria Energy Transition Plan and the National Renewable Energy and Energy Efficiency Policy.

Key lessons from this study include the importance of combining both supply- and demand-side interventions, the limitations of relying solely on nuclear energy (as shown by the poor performance of Scenario 6), and the critical role of co-benefits in strengthening the case for climate action. However, the study has limitations: it does not incorporate detailed cost-benefit analysis, investment requirements, or socio-political constraints that may affect implementation. Additionally, regional disparities and equity concerns were not explicitly modeled, though they remain central to a just energy transition.

The NECAL 2050 model proves to be a valuable tool for visualizing feasible low-carbon development pathways. Its results suggest that with coordinated policy efforts, stakeholder engagement, and targeted investments, Nigeria can achieve significant emission reductions without sacrificing energy access or economic growth. Future studies should expand on this work by integrating economic modeling, localized data, and stakeholder input to support the design of a more inclusive and resilient energy future.

Acknowledgements The authors are grateful to anonymous reviewers for their insightful suggestions.

Author contributions S.B: Conceptualisation, Data Curation, Formal Analysis, Investigation, Project Administration, Validation, Writing (Original Draft). M.M.O: Data Curation, Formal Analysis, Investigation, Validation, Supervision, Writing (Review & Editing).

Funding No funding was received for this study.

Data availability All data generated or analyzed during this study are included in this article.

Declarations

Ethics approval and consent to participate The authors declare that they have adhered to the ethical standards of research.

Consent for publication The authors declare their consent for publication.

Competing interests The authors declare no competing interests.

Human and animal rights This study does not contain any studies with human or animal subjects performed by any of the authors.

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